



FINAL PROJECT DATA SCIENCE TRAVEL

EMMA TRIVINI BELLINI

PROJECT OVERVIEW



This project involves working for an airline that is deeply committed to both providing high-quality service and promoting environmental sustainability. The main goal is to analyze passenger satisfaction data in order to identify areas of improvement and optimize services both during flights and on the ground.

 DATASETS: [ACCESS HERE](#)

INDEX

EXPLORATORY DATA ANALYSIS

-  DATASET EXPLORATION
-  MISSING VALUES
-  ENCODING OF CATEGORICAL VARIABLES
-  CHECKING FOR IMBALANCE
-  OUTLIERS
-  CORRELATION MATRIX
-  FEATURES SELECTION

EVALUATION METRICS FOR CLASSIFICATION MODELS

MODEL

-  WAY ONE
-  WAY TWO

CONCLUSIONS

1. EXPLORATORY DATA ANALYSIS <EDA>



✈️ GOAL: UNDERSTAND PASSENGER SATISFACTION DATASET, DETECT PATTERNS, HANDLE MISSING VALUES, AND PREPARE DATA FOR MODELING.

📁 1.1 DATASET OVERVIEW

- 103,904 ROWS, 25 COLUMNS IN TRAINING SET
- 25,976 ROWS, 25 COLUMNS IN TEST SET
- CONTAINS SURVEY INFO: GENDER, CUSTOMER TYPE, AGE, TRAVEL TYPE, CLASS, FLIGHT DISTANCE, AND MULTIPLE IN-FLIGHT SERVICE RATINGS
- TARGET VARIABLE: SATISFACTION → NEUTRAL/DISSATISFIED (0) OR SATISFIED (1)

❓ 1.2 MISSING VALUES

- ARRIVAL DELAY IN MINUTES HAD ~0.3% MISSING VALUES
- DECISION: REPLACE MISSING VALUES WITH 0 (TOLERABLE DATA LOSS)

1. EXPLORATORY DATA ANALYSIS <EDA>



🔄 1.3 ENCODING CATEGORICAL VARIABLES

- CATEGORICAL FEATURES: GENDER, CUSTOMER TYPE, TYPE OF TRAVEL, CLASS
- APPLIED ONE-HOT ENCODING
- TARGET VARIABLE MAPPED TO BINARY {0/1}
- DATASET READY FOR ANALYSIS

⚖️ 1.4 CLASS BALANCE

- SATISFIED VS NEUTRAL/DISSATISFIED: FAIRLY BALANCED
- NO RESAMPLING NEEDED

⚠️ 1.5 OUTLIERS

- REMOVED OUTLIERS USING IQR METHOD
- REDUCED TRAINING SET FROM 103,904 → 56,550 ROWS

1. EXPLORATORY DATA ANALYSIS <EDA>



1.6 CORRELATION ANALYSIS

- POSITIVE CORRELATION WITH TARGET:
 - ONLINE BOARDING, INFLIGHT ENTERTAINMENT, SEAT COMFORT, LEG ROOM SERVICE, CLEANLINESS
- NEGATIVE CORRELATION:
 - TYPE OF TRAVEL, CLASS (ECO/ECO PLUS)
- STRONG FEATURE CORRELATIONS:
 - TYPE OF TRAVEL ↔ CLASS (0.73)
 - INFLIGHT WIFI ↔ EASE OF ONLINE BOOKING (0.71)



1.7 FEATURE SELECTION

METHODS USED:

1. CHI-SQUARE TEST: FLIGHT DISTANCE, ONLINE BOARDING, TYPE OF TRAVEL, CLASS, INFLIGHT ENTERTAINMENT → MOST IMPORTANT
 2. MUTUAL INFORMATION: TYPE OF TRAVEL, ONLINE BOARDING, CLASS, INFLIGHT WIFI → MOST IMPORTANT
 3. T-TEST: ONLINE BOARDING, SEAT COMFORT, INFLIGHT ENTERTAINMENT, TYPE OF TRAVEL, CLASS → SIGNIFICANT IMPACT
- LOW IMPACT FEATURES: GENDER, GATE LOCATION, CUSTOMER TYPE → CAN BE IGNORED

2. EVALUATION METRICS FOR CLASSIFICATION MODELS



TO EVALUATE A CLASSIFICATION MODEL, WE NEED MORE THAN JUST ACCURACY. ACCURACY GIVES THE OVERALL PERCENTAGE OF CORRECT PREDICTIONS, BUT IT MAY BE MISLEADING IF THE DATA IS IMBALANCED. PRECISION INDICATES HOW MANY OF THE PREDICTED POSITIVES ARE TRULY CORRECT, WHILE RECALL MEASURES HOW MANY REAL POSITIVES ARE SUCCESSFULLY IDENTIFIED. THE F1-SCORE BALANCES PRECISION AND RECALL, MAKING IT USEFUL WHEN BOTH FALSE POSITIVES AND FALSE NEGATIVES ARE IMPORTANT.

OTHER METRICS PROVIDE ADDITIONAL PERSPECTIVES:

- AUC-ROC SHOWS HOW WELL THE MODEL SEPARATES CLASSES ACROSS THRESHOLDS.
- THE CONFUSION MATRIX DETAILS CORRECT AND INCORRECT CLASSIFICATIONS.
- LOG LOSS EVALUATES THE QUALITY OF PREDICTED PROBABILITIES.
- MCC AND COHEN'S KAPPA OFFER MORE ADVANCED EVALUATIONS, USEFUL FOR IMBALANCED DATASETS OR WHEN RANDOM AGREEMENT MUST BE CONSIDERED.

👉 THE CHOICE OF METRIC DEPENDS ON THE CONTEXT: SOMETIMES CAPTURING ALL POSITIVES IS THE PRIORITY, OTHER TIMES AVOIDING FALSE ALARMS IS MORE CRITICAL.

3. MODEL



TWO APPROACHES FOR BUILDING THE MODEL:

USE ALL AVAILABLE FEATURES.

USE ONLY THE MOST RELEVANT FEATURES IDENTIFIED IN THE T-TEST.

MODELS CHOSEN FOR BINARY CLASSIFICATION:

- RANDOM FOREST
- ADABOOST
- K-NEIGHBORS

EVALUATION WORKFLOW:

- SPOT CHECK WITH K-FOLD CROSS-VALIDATION TO COMPARE MODELS.
- HYPERPARAMETER TUNING USING RANDOMIZEDSEARCHCV ON THE BEST MODELS.

3. MODEL



3.1 WAY ONE (ALL FEATURES)

SPOT CHECK:

- RANDOM FOREST: ACCURACY_MEAN = 0.968, ACCURACY_STD = 0.0012
- K-NEIGHBORS: ACCURACY_MEAN = 0.934, ACCURACY_STD = 0.0006
- ADABOOST: ACCURACY_MEAN = 0.927, ACCURACY_STD = 0.0017

✓ RANDOM FOREST HAS THE BEST MEAN ACCURACY AND VERY LOW STD → STABLE ACROSS FOLDS.

HYPERPARAMETER TUNING:

- KNN: OPTIMIZED NEIGHBORS, WEIGHTS, METRIC → ACCURACY ON TEST SET = 0.892
- RANDOM FOREST: OPTIMIZED N_ESTIMATORS, MAX_DEPTH, MIN_SAMPLES → ACCURACY ON TEST SET = 0.896

OBSERVATION: SLIGHT DISCREPANCY ON TEST SET, BUT BOTH MODELS PERFORM WELL.

3. MODEL



3.2 WAY TWO (FEATURE SELECTION)

SELECTED FEATURES BASED ON T-TEST:

- TYPE OF TRAVEL
- ONLINE BOARDING
- INFLIGHT ENTERTAINMENT
- CLASS

SPOT CHECK:

- RANDOM FOREST: ACCURACY_MEAN = 0.8965, ACCURACY_STD = 0.0027
- ADABOOST: ACCURACY_MEAN = 0.8776, ACCURACY_STD = 0.0022
- K-NEIGHBORS: ACCURACY_MEAN = 0.8734, ACCURACY_STD = 0.0086

✓ RANDOM FOREST PERFORMS BEST AND IS STABLE, KNN SHOWS MORE VARIABILITY.

3. MODEL



3.2.2 HYPERPARAMETER TUNING (SELECTED FEATURES)

ADABOOST:

- BEST PARAMETERS: `N_ESTIMATORS = 50`, `LEARNING_RATE = 0.5`
- TEST SET ACCURACY = 0.823

RANDOM FOREST:

- BEST PARAMETERS: `N_ESTIMATORS = 100`, `MIN_SAMPLES_SPLIT = 2`, `MIN_SAMPLES_LEAF = 2`, `MAX_DEPTH = 30`
- TEST SET ACCURACY = 0.826

OBSERVATION: RANDOM FOREST CONSISTENTLY PERFORMS BEST, EVEN WITH FEWER FEATURES.

SUMMARY – KEY TAKEAWAYS

- RANDOM FOREST IS THE STRONGEST PERFORMER IN BOTH APPROACHES.
- FEATURE SELECTION SLIGHTLY REDUCES PERFORMANCE BUT SIMPLIFIES THE MODEL.
- KNN CAN SOMETIMES OUTPERFORM ON SPECIFIC TEST DATA, BUT SHOWS HIGHER VARIABILITY.
- HYPERPARAMETER TUNING IMPROVES MODEL ACCURACY AND STABILITY.

4.



CONCLUSIONS

MODEL PERFORMANCE OVERVIEW:

BASELINE (ALL FEATURES, NO TUNING):

- RANDOM FOREST: 96.8% ✓
- K-NEIGHBORS: 93.4%
- ADABOOST: 92.7%

WAY ONE (ALL FEATURES + HYPERPARAMETER TUNING):

- RANDOM FOREST: 89.79%
- K-NEIGHBORS: 89.16%
- HYPERPARAMETER TUNING IMPROVED PREDICTIVE ACCURACY FOR BOTH MODELS.

WAY TWO (SELECTED FEATURES):

- RANDOM FOREST: 82.61%
- ADABOOST: 82.28%
- ACCURACY DECREASED → FEATURE REDUCTION MAY HAVE REMOVED IMPORTANT INFORMATION.



4.



CONCLUSIONS



ANALYZING THE MODELS REVEALED KEY INSIGHTS INTO THEIR BEHAVIOR. RANDOM FOREST PROVED PARTICULARLY EFFECTIVE, HANDLING COMPLEXITY AND VARIABLE INTERACTIONS WELL.

HYPERPARAMETER TUNING IMPROVED PERFORMANCE FOR BOTH RANDOM FOREST AND K-NEIGHBORS, ENHANCING PREDICTIVE CAPABILITIES.

HOWEVER, REDUCING FEATURES LED TO A DECLINE IN ACCURACY, SHOWING THAT CAREFUL SELECTION IS ESSENTIAL TO AVOID REMOVING CRITICAL INFORMATION.

OVERALL, THE BEST PERFORMING MODEL IS RANDOM FOREST WITH TUNED HYPERPARAMETERS, BALANCING ACCURACY AND STABILITY.



THANK YOU FOR YOUR ATTENTION



PYTHON NOTEBOOK: [OPEN HERE](#)

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